Finite Permutation Groups

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1 Multiply transitive groups

Theorem 1.1. *Let Ω be a finite set and $G \leq \mathrm{Sym}(\Omega)$ be 2–transitive. Let $N \trianglelefteq G$ be a minimal normal subgroup. Then one of the following holds:*

(a) N is regular and elementary abelian.

(b) N is primitive, simple and not abelian.

Proof. First we show that N is unique. Suppose that M is another minimal normal subgroup of G, so $N \cap M = \{e\}$ and therefore $[N, M] = \{e\}$. Since non–trivial normal subgroups of the primitive group G are transitive, we obtain that both M and N are transitive. Together with Lemma ??? and $M \leq C_G(N)$ and $N \leq C_G(M)$ we get that both M and N are regular. By Theorem ???, regular normal subgroups of finite 2–transitive groups are abelian. Hence $<M, N>$ is abelian. But transitive abelian groups are regular, which forces $M = <M, N> = N$.

If N is regular, then we are done by Theorem ???.

So from now on we assume that N is not regular. Pick $\omega \in \Omega$. As G_ω permutes transitively the N_ω–orbits on $\Omega \setminus \{\omega\}$, we see that these N_ω–orbits all have the same length. By Wielandt's Lemma ???, we see that N is primitive or a Frobenius group.

We start to analyze the case that N is a Frobenius group. Let $F^\star$ be the set of fixed point free elements of N. By Lemma ???, $|F^\star| = n - 1$, where $n = |\Omega|$. Set

$$M = \{(\alpha, \beta, f) \mid \alpha, \beta \in \Omega, f \in F^\star, \beta = \alpha^f\}.$$

Note that for given f, there are n choices for α, and β then is unique. So $|M| = n(n-1)$. There is a natural action of G on M, for if $\beta = \alpha^f$ and $g \in G$, then $\beta^g = \alpha^{fg} = \alpha^{gg^{-1}fg} = (\alpha^g)^{f^g}$, so $(\alpha, \beta, f) \in M$ implies $(\alpha^g, \beta^g, f^g) \in M$. By the 2–transitivity of G, we see that for any $\alpha \neq \beta$ there is at least one $f \in F^\star$ with $(\alpha, \beta, f) \in M$. As there are $n(n-1) = |M|$ such pairs, we see that f is uniquely given by the pair α, β, and that the elements of $F^\star$ are conjugate. Let p be a prime divisor of Ω, and $f \in N$ be a element of order p. As f has either 0 or at least $p > 1$ fixed points, we conclude that $f \in F^\star$. So all elements of $F^\star$ have order p, hence Ω is a power of p. By Lemma ???, the Frobenius kernel $F = F^\star \cup \{e\}$ then is a subgroup of N. Let $g \in G$. Then $F^g \leq N$, and each element $\neq e$ in F^g is fixed point free, hence $F^g \subseteq F$ and therefore $F^g = F$. So F is a normal subgroup of G, and F is regular, because $|F| = |\Omega|$. But we assumed that G has no regular normal subgroup.

So we are left with the case that N is primitive and not regular. We need to show that N is simple. Suppose that this is not the case. Then N has a minimal normal subgroups $S < N$. As N is a minimal normal subgroup of G, there is an element $g \in G$ with $S \neq S^g$. So S and S^g are minimal normal subgroups of N, hence $S \cap S^g = \{e\}$ and therefore $[S, S^g] = \{e\}$. Again, the primitivity of N implies that S and S^g are transitive, and as these groups centralize each other, they both are regular. We claim that $N = SS^g$. If that were not the case, then there were $h \in G$ with $S^h \not\leq SS^g$. Similarly as above, $[S^h, SS^g] = \{e\}$, so SS^g we regular too, which is absurd. Thus $N = SS^g \cong S \times S$, so $|N| = |\Omega|^2$. Let m be the common length of the N_ω–orbits on $\Omega \setminus \{\omega\}$. So m divides $|\Omega| - 1$. But m also divides $|N_\omega|$, which divides $|N| = |\Omega|^2$. However $|\Omega| - 1$ and $|\Omega|^2$ are relatively prime. This final contradiction proves the theorem. $\square$

2 Jordan sets

Definition 2.1. Let G act transitively on Ω. A subset $\Delta \subseteq \Omega$ with $|\Delta| \geq 2$ is called a *Jordan set*, if the pointwise stabilizer of $\Omega \setminus \Delta$ in G acts transitively

on Δ.

Remark 2.2. If G is k–fold transitive Ω, then any subset Δ with $|\Delta| \geq 2$ and $|\Omega \setminus \Delta| \leq k-1$ is a Jordan set.

In the following we denote by $G(\Delta)$ the pointwise stabilizer of $\Omega \setminus \Delta$ in Δ.

Lemma 2.3. *Let G act transitively on Ω, and let Δ and Σ be Jordan sets of Ω. Then the following holds:*

(a) If $\Delta \cap \Sigma \neq \emptyset$, then $\Delta \cup \Sigma$ is a Jordan set.

(b) Let $\Delta \subsetneq \Omega$ be a maximal Jordan set with respect to inclusion. Then $\Delta \cap \Sigma = \emptyset$ or $\Sigma \subseteq \Delta$ or $\Delta \cup \Sigma = \Omega$.

(c) Let G be imprimitive, and B be a nontrivial block which intersects Δ non-trivially. Then $\Delta \subseteq B$ or $B \subseteq \Delta$.

Proof. (a) Clearly $<G(\Delta), G(\Sigma)>$ fixes $\Omega \setminus (\Delta \cup \Sigma)$ pointwise. As $G(\Delta)$ and $G(\Sigma)$ are transitive on Δ and Σ, and $\Delta \cap \Sigma$ is not empty, we get that $<G(\Delta), G(\Sigma)>$ is transitive on $\Delta \cup \Sigma$.

(b) Clear by (a).

(c) Suppose that the assertion is false. Then there is

$$\begin{aligned} \alpha &\in B \cap \Delta \\ \beta &\in B \setminus \Delta \\ \delta &\in \Delta \setminus B. \end{aligned}$$

Pick $g \in G(\Delta)$ with $\alpha^g = \beta$. Note that $\delta =\notin B$, but $\delta = \alpha^g \in B^g$, so $B \neq B^g$ and hence $B \cap B^g = \emptyset$. However, $\beta \notin \Delta$, so $\beta = \beta^g$ since $g \in G(\Delta)$. Therefore $\beta \in B \cap B^g$, a contradiction.

□

Lemma 2.4 (Rudio)**.** *Let G act primitively on the finite set Ω, and let Δ be a nonempty proper subset of Ω. Then for any distinct $\alpha, beta \in \Omega$ there is $g \in G$ with $\alpha \notin \Delta^g$, but $\beta \in \Delta^g$.*

Proof. Without loss of generality we may and do assume that G acts faithfully. In particular, G is finite.

Set $U := \{g \in G | \beta \in \Delta^g\}$ and $B = \cap_{g \in U} \Delta^g$. We are done once we know that B is a block, since then $B = \{\beta\}$ by primitivity of G, so there is $g \in U$ with $\alpha \notin \Delta^g$.

We need to show that B is a block. Let $x \in G$ be arbitrary, and assume that B and B^x have a nontrivial intersection. Pick $b \in B \cap B^x$. We need to show that $B = B^x$. We distinguish two cases:

(a) If $\beta \in B^x$, then $\beta \in \cap_{g \in U} \Delta^{gx}$, so $Ux \subset U$. But U is a finite set, hence $Ux = U$. We obtain

$$B^x = \cap_{g \in U} \Delta^{gx} = \cap_{h \in Ux} \Delta^h = \cap_{h \in U} \Delta^h = B.$$

(b) Now suppose that $\beta \notin B^x$. Pick $y \in G$ with $b^y = \beta$. From $b \in B \cap B^x$ we get $\beta = b^y \in B^y \cap B^{xy}$, so $\beta \in B \cap B^y \cap B^{xy}$. Now $B = B^y$ and $B = B^{xy}$ by case (a), hence $B^y = B^{xy}$, and finally $B = B^x$.

□

Theorem 2.5 (Jordan)**.** *Let $G \leq \mathrm{Sym}(\Omega)$ act primitively on the finite set Ω. Suppose that there is a proper Jordan set in Ω. Then the action is 2–transitive.*

Proof. Let $\Delta \subsetneq \Omega$ be a maximal nontrivial Jordan set. Set $|\Delta| = k$ and $|\Omega| = n$. Fix $\alpha \in \Omega$. By Rudio's Lemma, for any $\beta \in \Omega \setminus \{\alpha\}$ there is $g \in G$ with $\alpha \notin \Delta^g$, but $\beta \in \Delta^g$. Clearly, Δ^g is a proper Jordan set too. Let D be the set of these sets Δ^g. Note that any two members of D intersect trivially by Lemma ???. So $\Omega \setminus \{\alpha\}$ is a disjoint union of the sets in D. In particular, k divides $n-1$. Set

$$M = \{(\alpha, \Delta^x) | \alpha \in \Omega, x \in G, \alpha \notin \Delta^x\}.$$

Counting via α shows $|M| = n \cdot \frac{n-1}{k}$.

Set $u := |\{\Delta^x | x \in G\}|$. Clearly $|M| = u \cdot (n-k)$, so

$$uk(n-k) = n(n-1).$$

From $k \mid n-1$ we get that the numbers k, $n-k$ and n are pairwise relatively prime, so $k(n-k)$ divides $n-1$. In particular, $k(n-k) \leq n-1$. Together with $2 \leq k \leq n-1$ this implies $k = n-1$, so $\Omega = \Delta \cup \{\alpha\}$ for some α. We get that G_α is transitive on Δ, and the claim follows. □

3 Sharply multiply transitive groups

Let G act on the set Ω. Set $\Omega^{(k)}$ be the set of k–tuples $(\omega_1, \omega_2 \ldots, \omega_k)$ with pairwise distinct entries. We say that G is *sharply k–fold transitive* on Ω if G acts regularly on $\Omega^{(k)}$.

Some easy consequences of this definition are:

- If G is transitive on Ω, then G is sharply k–fold transitive on G if and only G_ω is sharply $(k-1)$–fold transitive on $\Omega \setminus \{\omega\}$ for some (or any) $\omega \in \Omega$.

- If G is sharply k–fold transitive on the finite set Ω of size n, then $|G| = n(n-1)(n-2)\dots(n-k+1)$.

Clearly, a sharply 1–fold transitive group is just a group in its regular action. We list some examples of sharply k–fold transitive groups for $k \geq 2$:

- Let K be a skewfield. Then $G = \mathrm{AGL}(1, K)$ acts sharply 2–transitively on K.

- Let K be field. Then $\mathrm{PGL}(2, K)$ acts sharply 3–transitively on the set of 1–dimensional subspaces of K^2.

- If $|\Omega| = n < \infty$, then $\mathrm{Sym}(\Omega)$ acts sharply n–transitively on Ω. This action is also sharply $(n-1)$–transitive. Furthermore, $\mathrm{Alt}(\Omega)$ acts sharply $(n-2)$–transitively on Ω.

A Theorem of Tits shows that except for specific groups of degrees 11 and 12, there are no other cases of sharp k–transitivity for $k \geq 4$.

In the following M_{11} denotes a hypothetical group which acts sharply 4–transitively on 11 points. Such a group indeed exists and is unique, which will not be shown here. (Remarks about Mathieu groups ???)

In the following we do not assume that Ω is finite. Before discussing the case $k \geq 4$, we look at the case $k = 4$ which serves as basis for an induction proof of the general case.

Proposition 3.1 (Tits)**.** *Let G act sharply 4–transitively on the set Ω. Then G is one of the following groups: S_4, S_5, A_6, M_{11}.*

Proof. We will frequently show that an element from G is the identity by showing that it has at least four fixed points.

Let $1, 2, 3, 4$ be distinct elements from Ω. Pick $t \in G$ with

$$1^t = 2,\ 2^t = 1,\ 3^t = 3,\ 4^t = 4.$$

So t^2 has at least four fixed points, hence $t^2 = e$.

So t is an involution. Set $H = G_1 \cap C_G(t)$. Clearly, H acts on the fixed points of t. This action is faithful, because $h \in H$ fixes 1, and also 2 in view of $2^h = 1^{th} = 1^{ht} = 1^t = 2$. So if h fixes 3 and 4 too, then h has four fixed points, hence $h = e$.

On the other hand, t has at most 3 fixed points, and if $|\Omega|$ is even, then t has exactly 2 fixed points. From that we see that $|H| \leq 6$, and $|H| \leq 2$ if $|\Omega|$ is even.

Let

$$P = \{(\omega, \omega^t) | \omega \in \Omega, \omega \neq 1, 2, \omega \neq \omega^t\}$$

be the ordered orbits of $<t>$ of length 2 distinct from $(1, 2)$ and $(2, 1)$.

We claim that H acts transitively on P: Pick (ω, ω^t) and $(\bar{\omega}, \bar{\omega}^t)$ in P, and choose $g \in G$ with $1^g = 1$, $2^g = 2$, $\omega^g = \bar{\omega}$, $(\omega^t)^g = \bar{\omega}^t$.

Of course $g \in G_1$. We are done once we know that $g \in C_G(t)$. Consider the commutator $c = g^{-1}t^{-1}gt$, and note that $t = t^{-1}$. We compute that c fixes the points $1, 2, \bar{\omega}, \bar{\omega}^t$:

$$\begin{array}{ccccccccc}
1^{g^{-1}t^{-1}gt} & = & 1^{t^{-1}gt} & = & 2^{gt} & = & 2^t & = & 1 \\
2^{g^{-1}t^{-1}gt} & = & 2^{t^{-1}gt} & = & 1^{gt} & = & 1^t & = & 2 \\
\bar{\omega}^{g^{-1}t^{-1}gt} & = & \omega^{t^{-1}gt} & = & (\omega^t)^{gt} & = & (\bar{\omega}^t)^t & = & \bar{\omega} \\
(\bar{\omega}^t)^{g^{-1}t^{-1}gt} & = & (\omega^t)^{t^{-1}gt} & = & \omega^{gt} & & & = & \bar{\omega}^t
\end{array}$$

This yields $g^{-1}t^{-1}g = e$, so $g \in H$.

Note that

$$|\Omega| = 2 + f + |P|,$$

where $f \in \{2, 3\}$ is the number of fixed points of t. We obtain $|\Omega| \leq 2+3+6 = 11$, and $|\Omega| \leq 2 + 2 + 2 = 6$ if $|\Omega|$ is even.

In order to finish the proof, we need to show that $|\Omega| \neq 7, 9$.

First suppose that $|\Omega| = 9$. Then $|P| = 4$. However, H is isomorphic to a subgroup of S_3, and such a group cannot act transitively on a set of size 4.

So we are left to deal with $|\Omega| = 7$. In this case t is a double transposition. But G is 4–transitive on Ω, so G then contains all double transpositions from $\mathrm{Sym}(\Omega)$. But $(1\ 2)(3\ 4) \cdot (1\ 2)(3\ 5) = (3\ 4\ 5)$, so G also contains all 3–cycles, hence $G \geq \mathrm{Alt}(\Omega)$, a contradiction. □

Lemma 3.2. *Let $G \leq \mathrm{Sym}(\Omega)$ be transitive, and $|\Omega| = p$ be a prime. Then $|G| = pd(1 + pe)$ with $e \in \mathbb{N}_0$ and a divisor d of $p - 1$.*

Proof. Let C be a Sylow p–subgroup of G. Then C has order p, is cyclic and regular on Ω. Set $N = N_G(C)$. Note that $C = C_G(C)$, so N/C is isomorphic to a subgroup of $\mathrm{Aut}(C)$, a group of order $d \mid p - 1$. The number of Sylow p–subgroups of G has the form $1 + pe$, and equals $[G : N]$ since they are all conjugate to C. The claim then follows from $|G| = [G : N][N : C]|C|$. □

Lemma 3.3. *Suppose that $|\Omega| = 13$. Then $\mathrm{Sym}(\Omega)$ does not contain a sharply 6–transitive group.*

Proof. Suppose that $G \leq \mathrm{Sym}(\Omega)$ is sharply 6–transitive. So $|G| = 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8$. By the previous lemma, $|G| = 13d(1+13e)$ for a divisor d of 12. Writing $12 = dk$, we obtain $11 \cdot 10 \cdot 9 \cdot 8k = 1 + 13e$, hence $3k \equiv 1 \pmod{13}$. This yields $k \equiv 9 \pmod{13}$, contrary to k being a divisor of 12. □

In the following M_{12} denotes a group which acts sharply 5–transitively on 12 points. We do not prove existence and uniqueness of such a group.

Theorem 3.4 (Tits)**.** *Suppose that G act sharply k–transitively on the set Ω for $k \geq 4$. Then Ω is finite, and G is one of the following groups: S_n ($k = n$ or $k = n-1$), A_n ($k = n-2$), M_{11} ($k = 4$), M_{12} ($k = 5$).*

Proof. We know this result already for $k = 4$. Since a point stabilizer of a sharply k–transitive group is sharply $(k-1)$–transitive on the remaining points, the result follows by induction and the fact from the previous lemma, that M_{12} is not a point stabilizer of a sharply 6–transitive group on 13 elements. □

4 Transitive groups of prime degree

Theorem 4.1 (Burnside)**.** *Let p be a prime number, $|\Omega| = p$, and $G \leq \mathrm{Sym}(\Omega)$ be transitive. Then G is either 2–transitive or a proper subgroup of $\mathrm{AGL}_1(p)$ with respect to a suitable identification of Ω with $\mathbb{F}_p$.*

The main tool is the following

Proposition 4.2. *Let U be a non-empty, proper subset of $\mathbb{F}_p \setminus \{0\}$. Let π be a permutation of $\mathbb{F}_p$ such that $i - j \in U$ for $i, j \in \mathbb{F}_p$ implies $\pi(i) - \pi(j) \in U$. Then there are $a, b \in \mathbb{F}_p$ such that $\pi(x) = ax + b$ for all $x \in \mathbb{F}_p$.*

In [Sch08] Schur gives a proof of this proposition in two steps. First he uses a precursor of his S-ring technique to show that if $1 \in U$, then U is a subgroup of $\mathbb{F}_p \setminus \{0\}$. In the second step he shows that π is linear. In this note we show that a small modification of his second step makes the first step unnecessary. See the remarks at the end for further comments.

Proof of Burnside's theorem. Let G be a transitive permutation group on p elements. As p divides the order of G, there is an element $\tau \in G$ of order p. We may assume that G acts on $\mathbb{F}_p$, with $\tau(x) = x + 1$ for all $x \in \mathbb{F}_p$. So G contains the group of translations $x \mapsto x + b$ for each $b \in \mathbb{F}_p$.

Suppose that G is not doubly transitive. Then the stabilizer G_0 of 0 has at least 2 orbits on $\mathbb{F}_p \setminus \{0\}$. Let U be one of these orbits. In order to show

that U meets the assumptions of the proposition, pick i, j with $i - j \in U$. Define $\sigma : x \mapsto \pi(x + j) - \pi(j)$, note that $\sigma \in G$, and $\sigma(0) = 0$, so actually $\sigma \in G_0$. Therefore $\pi(i) - \pi(j) = \sigma(i - j) \in U$.

So we may apply the proposition, which shows that G is a subgroup of the group of permutations $x \mapsto ax + b$ with $a \in \mathbb{F}_p \setminus \{0\}, b \in \mathbb{F}_p$.

Lemma 4.3. *Let U be a subset of m nonzero elements in a field. Then there is an integer r with $1 \le r \le m$ with $\sum_{u \in U}^m u^r \neq 0$.*

Proof. Write $U = \{u_1, u_2, \ldots, u_m\}$ and $\prod_{i=1}^m (X - u_i) = a_0 + a_1 X + \cdots + a_{m-1} X^{m-1} + X^m$. Consider the matrix and the vectors

$$A = \begin{pmatrix} 1 & u_1 & \cdots & u_1^{m-1} \\ 1 & u_2 & \cdots & u_2^{m-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & u_m & \cdots & u_m^{m-1} \end{pmatrix}, \; a = \begin{pmatrix} a_0^m \\ a_1^m \\ \vdots \\ a_{m-1} \end{pmatrix}, \; v = -\begin{pmatrix} u_1^m \\ u_2^m \\ \vdots \\ u_m^m \end{pmatrix}.$$

Then a is the unique solution of the linear equation $Ax = v$. Therefore A is invertible. In particular

$$\begin{pmatrix} \sum_{u \in U} u & \sum_{u \in U} u^2 & \cdots & \sum_{u \in U} u^m \end{pmatrix} = \begin{pmatrix} u_1 & u_2 & \cdots & u_m \end{pmatrix} A \neq 0,$$

and the claim follows. □

Proof of the proposition. By an iterated application of π we see that $i - j \in U$ if and only if $\pi(i) - \pi(j) \in U$. In particular, replacing U by its complement in $\mathbb{F}_p \setminus \{0\}$ preserves the assumption. Therefore we may and do assume $|U| \le \frac{p-1}{2}$.

Fix $i \in \mathbb{F}_p$. For $u \in U$ we have $(i + u) - i \in U$, hence $\pi(i + u) - \pi(i) \in U$. As π is a permutation, the elements $\pi(i+u) - \pi(i)$ are different for different u. Thus $\{\pi(i + u) - \pi(i) | \, u \in U\} = U$, hence $\{\pi(i + u) | \, u \in U\} = \{\pi(i) + u | \, u \in U\}$. In particular, for $w \in \mathbb{N}$ we obtain

$$\sum_{u \in U} \pi(i + u)^w = \sum_{u \in U} (\pi(i) + u)^w.$$

Let $f(X) \in \mathbb{F}_p[X]$ be the polynomial of degree $n \le p - 1$ with $f(i) = \pi(i)$ for all $i \in \mathbb{F}_p$. Suppose $wn \le p - 1$. Then $\sum_{u \in U} f(X + u)^w - \sum_{u \in U} (f(X) + u)^w$ is a polynomial of degree $< p$ which vanishes identically on $\mathbb{F}_p$, thus

$$\sum_{u \in U} f(X + u)^w - \sum_{u \in U} (f(X) + u)^w = 0.$$

Setting $S_k = \sum_{u \in U} u^k$, we obtain

$$\sum_{u \in U} (f(X+u)^w - f(X)^w) = \sum_{k=1}^{w} \binom{w}{k} S_k f(X)^{w-k}.$$

Let $r \geq 1$ be minimal with $S_r \neq 0$. Then the right hand side has degree $n(w-r)$ if $r \leq w$, or vanishes if $r > w$.

$f(X)^w$ is a polynomial of degree $nw < p$. For $0 \leq \nu \leq nw$, the νth derivative $(f(X)^w)^{(\nu)}$ has degree $nw - \nu$. Thus there are $a_0, a_1, \ldots, a_{nw} \in \mathbb{F}_p$ such that

$$X^{nw} = \sum_{\nu=0}^{nw} a_\nu (f(X)^w)^{(\nu)},$$

where $a_0 \neq 0$. This yields

$$\sum_{u \in U} ((X+u)^{nw} - X^{nw}) = \sum_{\nu=0}^{nw} a_\nu \left(\sum_{k=1}^{w} \binom{w}{k} S_k f(X)^{w-k} \right)^{(\nu)}.$$

So

$$\sum_{u \in U} ((X+u)^{nw} - X^{nw}) \text{ is 0 or has degree } n(w-r). \tag{1}$$

The previous lemma gives $r \leq |U| \leq \frac{p-1}{2}$.

Suppose we have chosen w maximal with $nw \leq p-1$. Then $p-1 < n(w+1) \leq 2nw$, so $nw > \frac{p-1}{2} \geq r$. Thus $nw - r > 0$, and (1) has degree $nw - r$ as one sees from expanding $(X+u)^{nw}$. We obtain $nw - r = n(w-r)$, so $n = 1$, and we are done.

Remark. Our proof resembles the final step of Schur's proof in [Sch08]. However, the main part of his proof consists in showing that if $1 \in U$, then U is a subgroup of $\mathbb{F}_p \setminus \{0\}$. Thus if $1 \leq k \leq w < |U|$, then $S_k = 0$, so $\sum_{u \in U} f(X+u)^w = |U| f(X)^w$, which produces a contradiction similarly as above. See also [DM96, 3.5] for a modern version of this proof.

In [DKM92] the authors give an S-ring argument to show that U is a group. From there they however proceed with geometric arguments, and use facts about lacunary polynomials to conclude that π is a linear function.

Burnside's original proof uses complex character theory, see [Bur11].

Another proof is due to Wielandt, who studies the ring of G-invariant functions from $\mathbb{F}_p$ to $\mathbb{F}_p$. See [Wie94, pages 273–296], [HB82, XII, §10]. Concise and streamlined version of Wielandt's proof are contained in [LMT93, 6.7] and [FJ05, 21.7].

5 Primitive groups with regular cyclic subgroup

We study finite primitive permutation groups with a cyclic regular subgroup. In a long and difficult paper from 1933 I. Schur [Sch33] invented a new technique to prove that such a group has either prime degree, or is doubly transitive. Later Wielandt [Wie64] gave a different version of Schur's technique and proved the slightly more general

Theorem 5.1 (Schur)**.** *Let G be a primitive permutation group of degree n, and A be a regular abelian subgroup. Suppose that n is not prime, and that A contains a cyclic Sylow p–subgroup for some prime divisor p of $|A|$. Then G is doubly transitive.*

Because of its importance for the monodromy groups of polynomials and the beauty of its proof, we give a self–contained account in the following. Our presentation is influenced by [DM96]. For a variant confer [LMT93].

In the following we assume the hypotheses of the Theorem. Let Ω be the set G acts on. Pick $\omega \in \Omega$. The regular action of A allows us to identify Ω with the elements of A, where we identify a with ω^a. This way we obtain a regular action of G on A, where the subgroup A acts in its regular action on itself. In order to avoid a confusion with the notation for conjugation, we write $a \star g$ for the image of $a \in A$ under $g \in G$ instead of a^g. If $b \in A$, then obviously $a \star b = ab$. Also,

$$1 \star (ag) = (1 \star a) \star g = a \star g = 1 \star (a \star g) \text{ for all } a \in A,\ g \in G. \tag{2}$$

Let p be a prime such that, according to the hypothesis, A has a cyclic p–Sylow subgroup. So A has a unique subgroup of order p, call it P.

We will work with the group algebra R of A over $\mathbb{F}_p$. As an $\mathbb{F}_p$–space, this is the space of mappings $\lambda : A \to \mathbb{F}_p$. Addition and multiplication in R are defined as follows:

$$\begin{aligned}(\lambda + \mu)(a) &:= \lambda(a) + \mu(a)\\ (\lambda\mu)(a) &:= \sum_{\substack{b,c \in A\\ bc=a}} \lambda(b)\mu(c)\end{aligned}$$

It is straightforward to verify that this makes R into a ring with unity. We use a more convenient notation, by writing λ_a instead of $\lambda(a)$ and by identifying $\lambda \in R$ with the formal linear combination $\sum_{a \in A} \lambda_a a$.

We extend the action of G on A to R by linearity, and again write $r \star g$ for $g \in G$ applied to $r \in R$.

For a subset $\Gamma \subseteq A$ we denote by $[\Gamma]$ the element $\sum_{\gamma\in\Gamma}\gamma \in R$. For $r \in R$, let $supp(r)$ be the support of $r = \sum \lambda_a a$, which is the set of those $a \in A$ with $\lambda_a \neq 0$. Let G_1 be the stabilizer of $1 \in A$, and denote by $\mathbf{C}_R(G_1)$ the subset of elements of R which are fixed by G_1. Let $A_1 = \{1\}, A_2, \ldots, A_N$ be the orbits of G_1 on A. Of course, we aim to prove that $N = 2$.

Lemma 5.2. *The following holds.*

(a) $\mathbf{C}_R(G_1)$ *is an* $\mathbb{F}_p$*–subspace of* R *with basis* $[A_1], [A_2], \ldots, [A_N]$.

(b) If $\Gamma \subseteq A$ *is* G_1*–invariant, then* $(\Gamma a) \star g = \Gamma a \star g$ *for all* $a \in A$, $g \in G_1$.

(c) If $\Gamma, \Delta \subseteq A$ *are* G_1*–invariant, then so is* $\Gamma\Delta$.

(d) $\mathbf{C}_R(G_1)$ *is a subring of* R.

(e) If B *is a* G_1*–invariant subgroup of* A*, then* $B = \{1\}$ *or* $B = A$.

(f) $<supp(c)> = 1$ *or* A *for each* $c \in \mathbf{C}_R(G_1)$.

(g) Let $\Gamma \subseteq A$ *be* G_1*–invariant. Then* $[\Gamma]^p = |\Gamma \cap P|1$.

(h) For each $i = 1, 2, \ldots, N$ *we have* $|A_i \cap P| \geq 1$ *and* $A_i \setminus P$ *is a union of cosets of* P *in* A.

Proof. To (a). If $c = \sum_{a\in A} \lambda_a a \in \mathbf{C}_R(G_1)$, then $c \star g = c$ for all $g \in G_1$ implies $\lambda_a = \lambda_b$ if a, b lie in the same G_1–orbit. From this conclude the claim.

To (b). From $1\star(ag) = 1\star(a\star g)$ (see (2)) we get $h \in G_1$ with $ag = h(a\star g)$. So, for $\gamma \in \Gamma$ we obtain from a repeated application of (2) and noting that $\gamma a, \gamma \star h, a \star g \in A$ the following:

$$\begin{aligned}(\gamma a) \star g &= 1 \star (\gamma a g)\\ &= 1 \star (\gamma h(a \star g))\\ &= 1 \star ((\gamma \star h)(a \star g))\\ &= (\gamma \star h)(a \star g).\end{aligned}$$

If γ runs through Γ, then so does $\gamma \star h$, and (b) follows.

To (c). Let a run through Δ and use (b).

To (d). In virtue of (a) we need to show that $[A_i][A_j] \in \mathbf{C}_R(G_1)$ for all i, j. For $g \in G_1$ we get from (b) that if $b \in A_j$, then $(A_i b) \star g = A_i(b \star g)$, hence $([A_i]b) \star g = [A_i](b \star g)$. Now take the sum over $b \in A_j$, and note that $b \star g$ runs through A_j.

To (e). If $h \in G_1$ then $B \star h = B$ by assumption. As $G = G_1 A$, any $g \in G$ has the form $g = ha$ with $h \in G_1$ and $a \in A$. So $B \star g = (B \star h) \star a = B \star a = Ba$. So the coset Ba either equals B, or is disjoint from B. We see that B is a block with respect to the action of G on A. However, this action is primitive, forcing $|B| = 1$ or $B = A$.

To (f). For $c \in \mathbf{C}_R(G_1)$ let S be the support of c, and B the group generated by S. Note that S is G_1–invariant. Then $B = \bigcup_{i=0}^{\infty} S^i$, and B is G_1–invariant by (c). The assertion follows from (e).

To (g). Clearly $[\Gamma] \in \mathbf{C}_R(G_1)$. As the support of $[\Gamma]^p$ consists of p-th powers of A, and the set of p-th powers of A is a proper subgroup of A, we see that the group generated by the support of $[\Gamma]^p$ is a proper subgroup of A, hence is trivial by (f). So $[\Gamma]^p$ is a multiple of 1, and the coefficient of 1 is the number of $\gamma \in \Gamma$ with $\gamma^p = 1$, and the claim follows.

To (h). Let Γ be one of the A_i. Pick $\gamma \in \Gamma \setminus P$. Then $\gamma^p \neq 1$, and the number of elements $\delta \in \Gamma$ with $\gamma^p = \delta^p$ must be divisible by p as a consequence of (g). So this number is at least p. On the other hand, as A is abelian, $\gamma^p = \delta^p$ implies $\delta \in P\gamma$, so there are at most $|P| = p$ such elements δ. It follows that the set of $\delta \in \Gamma$ with $\delta^p = \gamma^p$ is precisely $P\gamma$. In particular, $P\gamma \subseteq \Gamma$. We see that $\Gamma \setminus P$ is a union of cosets of P.

It remains to show that Γ contains an element from P. Suppose wrong. Then, by what we saw, Γ is a union of cosets of P, hence $P \subseteq B := \{b \in A |\ \Gamma b = \Gamma\}$. Clearly B is a subgroup of A of order > 1. Furthermore, B is G_1 invariant by (b). So $B = A$ by (f). But this implies the nonsense $A = \Gamma A = \Gamma \subsetneqq A$. $\square$

Now we are ready to prove the theorem. Suppose that G is not doubly transitive. Then there exists a G_1–orbit Γ different from $A_1 = \{1\}$ such that $\mu := |\Gamma \cap P| \leq (p-1)/2$. Set $S = \{a \in A | Pa \subseteq \Gamma\}$. By (h) we know that Γ is the disjoint union of $\Gamma \cap P$ and the sets Pa, $a \in S$. Taking sums, we get

$$[\Gamma] = [\Gamma \cap P] + [P]r,$$

with $r = [S]$. For $v \in P$ we have $v[P] = [P]$, hence $[P]^2 = p[P] = 0$ and $[\Gamma \cap P][P] = \mu[P]$. Now $c := ([\Gamma] - \mu 1)^2 \in \mathbf{C}_R(G_1)$ by (d). We compute

$$\begin{aligned} c &= ([\Gamma \cap P] - \mu 1 + [P]r)^2 \\ &= ([\Gamma \cap P] - \mu 1)^2 + 2([\Gamma \cap P][P] - \mu[P])r + [P]^2 r^2 \\ &= ([\Gamma \cap P] - \mu 1)^2, \end{aligned}$$

so the support of c is contained in P. Since $P < A$, we get $supp(c) \subseteq \{1\}$ by (f). So $([\Gamma \cap P] - \mu 1)^2 = [\Gamma \cap P]^2 - 2\mu[\Gamma \cap P] + \mu^2 1$ is a multiple of 1. By (h), there is $\gamma \in \Gamma \cap P$, so $\mu \geq 1$ and $\gamma \neq 1$. Let ρ be the number of appearances

of γ in the expansion of $[\Gamma \cap P]^2$. Clearly $\rho \leq \mu$. But by the above, p divides $\rho - 2\mu$, a contradiction to $-p < -2\mu \leq \rho - 2\mu \leq \mu - 2\mu < 0$.

Remark. There are also versions of the theorem for certain non-abelian regular groups A, see the references given in [Wie64, §25].

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